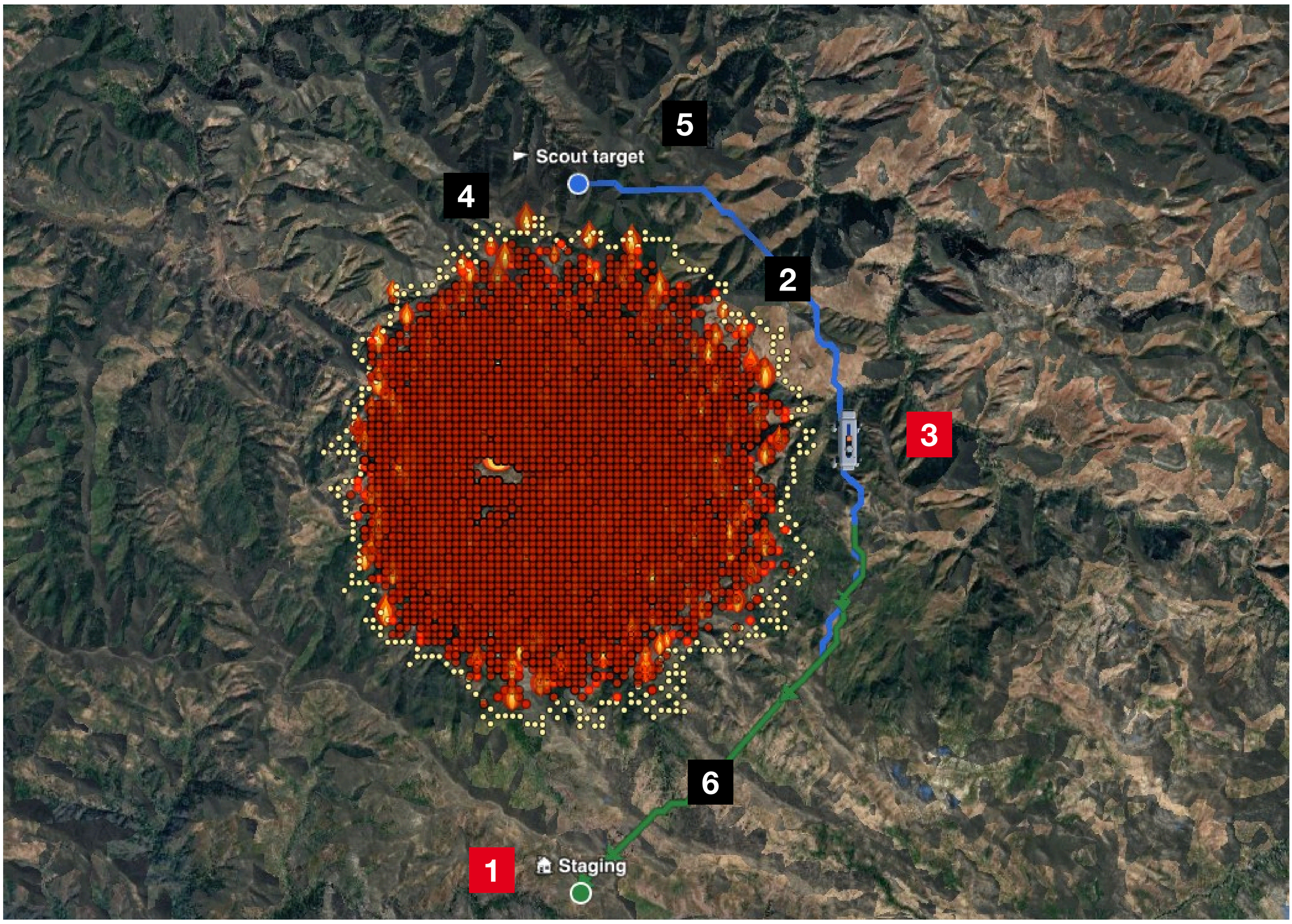
Fire-aware robot dogs

Machine-learning wildfire forecasts for safe navigation and scouting

Shawn Zeng
Monta Vista High School

One afternoon at a Big Sur fire

A simulated mission on real terrain. Follow the dog from stop 1 to stop 6. How it decides at each stop: right panel.



1 **3:00 PM** Launch
Steep, brushy slopes, burning since midnight. Too dangerous for a crew, so the dog leaves staging.

2 **3:00 PM** Plan
The **blue** route keeps 100 m from the forecast fire, with 30 min in hand, slower in brush and uphill.

3 **3:30 PM** Wind shift
The real fire turns; the forecast doesn't know, so the prediction needs a rerun. The **green** escape path stays open.

4 **4:45 PM** Scout
It reaches its scout spot, as close as the 100 m rule allows.

5 **4:45 PM** Measure
Its sensors find wind from the northwest, not the west, and 40 °C, not 35. The prediction reruns: the fire is coming, so the dog re-plans and heads home another way.

6 **~7 PM** Home safe
Back before the fire arrives. The crew plans with a better forecast.

Know where the fire will be, not just where it is.

Safer crews
Fewer firefighters sent into danger to look.

Robot stays safe
Routes and exits keep it from getting trapped.

Smarter sensing
Measure where it changes decisions.